

# ISHU SINGH

GIS Engineer | Geospatial Software Developer | Remote Sensing | AI Automation

---

**Email:** singhishu2060@gmail.com

**LinkedIn:** linkedin.com/in/ishu-singh-geo

**GitHub:** github.com/ishunteam-png

**Portfolio:** ishunteam-png.github.io

**GIS Work:** ishunteam-png.github.io/gis-portfolio (17 live dashboards)

**Location:** Tbilisi, Georgia — available for remote or relocation

---

## PROFESSIONAL SUMMARY

GIS engineer and geospatial software developer with three years of hands-on work in satellite remote sensing, SAR/InSAR analysis, spatial data pipelines, and full-stack web development. Self-taught the complete MintPy/SNAP InSAR workflow from scratch, ran a 3-year Sentinel-1 stack over Delhi road infrastructure, and produced an EGMS Level 3 equivalent output — 2,655 Persistent Scatterers, true 2D deformation decomposition, subsidence rates from -10.3 to +4.7 mm/yr.

On the software side: three production systems shipped and running — an AI-assisted web platform, a Telegram/n8n automation suite with 7 live workflows, and a healthcare platform now used by four hospitals. Also built 17 independent geospatial research dashboards covering LULC classification, flood risk, mobility analytics, wildfire modelling, crop type discrimination, air quality, vessel tracking, and urban heat mapping. Work at every layer from data ingestion and pipeline engineering to React front-end.

---

## TECHNICAL SKILLS

**Programming:** Python | TypeScript | JavaScript | React | HTML / CSS

**GIS & Remote Sensing:** GeoPandas | rasterio | GDAL | MintPy | SNAP | PyAPS3 | Google Earth Engine | Sentinel-1/2/5P | Landsat 8/9 | LandTrendr | TROPOMI | VIIRS

**Spatial Analysis:** OSMnx | NetworkX | Shapely | h3-py | pystac-client | xarray | OR-Tools | pvlib

**Machine Learning:** scikit-learn | PyTorch | YOLOv8 / ultralytics | OpenCV | DBSCAN | Random Forest | Sensor Fusion

**Data Engineering:** DuckDB | pandas | pyarrow | scipy | NumPy | Leaflet.js | Chroma.js

**Automation & DevOps:** n8n | Telegram Bot API | GitHub Actions | Vite | Tailwind CSS | GitHub Pages

---

## WORK EXPERIENCE

*Note: The section below covers paid, client-facing, and commercially deployed engineering work. Portfolio and research projects appear in the next section.*

### Freelance GIS Engineer & Remote Sensing Developer

Self-employed — client and research work | Remote

2022 – Present

- Built an end-to-end Sentinel-1 InSAR pipeline from scratch — SLC download, coregistration, SBAS network, atmospheric correction with PyAPS3, time-series inversion in MintPy, and 2D decomposition to vertical and east-west components. Output matched EGMS Level 3 specifications on a 3-year Delhi stack with 2,655 Persistent Scatterers at -10.3 to +4.7 mm/yr.
- Ran YOLOv8m semantic segmentation over road surface imagery, then spatially joined detections to the InSAR PS dataset — every damage detection landed on a subsiding scatterer, giving a ground-truth-backed risk flag for infrastructure maintenance teams.
- Classified Bengaluru land cover from a 2020-2024 Sentinel-2 time series (OA 90.9%, kappa 0.89) and fused Sentinel-1 SAR polarimetry with Sentinel-2 optical for low crop mapping — fusion pushed accuracy from 86.3% to 92.7%, a +6.4 percentage-point gain across 672 fields.
- Tracked 2,698 km<sup>2</sup> of Amazonian deforestation between 2015 and 2024 using LandTrendr breakpoint detection on Earth Engine. Mapped shoreline migration at 15 Florida stations over a decade: Daytona Beach -27.4 m net retreat, Miami Beach +24.0 m from federal renourishment.
- Modelled multi-signal café location suitability across a 62,849-node Tbilisi pedestrian network (Spearman rho = 0.39 vs real distribution); benchmarked OR-Tools GLS routing against greedy and Clarke-Wright — OR-Tools cut total drive time by 20.3% across 60 stops and 5 vehicles.

## Freelance Software Engineer & Automation Developer

Haimcore (AI platform) · Relief Guru (workflow automation) · Jalaram Healthcare | Remote

2022 – Present

- Shipped Haimcore — a production AI-assisted web platform. Built release-versioned audit folders, rollback paths for every deployment, and idempotency guards so repeated runs never corrupt state. Hard-fail at every LLM API boundary with the actual error, not a silent fallback.
- Built Relief Guru: Telegram-driven n8n automation with 7 workflows in production. Every external API call has typed error propagation. Deploy process is a single script with a smoke test and a rollback path. No manual steps.
- Developed and maintains Jalaram — a multi-tenant healthcare web app with 4 live hospital portals. Features include SystemLog audit trails for every clinical and admin event, role-based access control, and real-time patient management dashboards.

---

## INDEPENDENT GEOSPATIAL RESEARCH & PORTFOLIO PROJECTS

*These are self-directed research and learning projects, not client work. All 17 are live at [ishunteam-png.github.io/gis-portfolio](https://ishunteam-png.github.io/gis-portfolio) — each has a full interactive Leaflet dashboard and a GitHub repo with the analysis code.*

### InSAR Road Subsidence — Delhi

Sentinel-1 · MintPy · SNAP · PyAPS3

The flagship project. 3-year SAR stack, 2,555 PS identified, -10.3 to +4.7 mm/yr subsidence, EGMS-L3 equivalent output. Live dashboard with velocity threshold filter and temporal coherence gate.

### GeoAI Road Damage x InSAR — Delhi

YOLOv8m · PyTorch · OpenCV

Semantic segmentation of road damage photos joined to the InSAR PS layer. All 4 damage detections landed on subsiding scatterers — validates the satellite-to-infrastructure failure chain.

### Sentinel-2 LULC Classification — Bengaluru 2020–2024

pystac-client · scikit-learn · rasterio

OA 90.9%, kappa 0.89. Built-up area +22.6% over 4 years. Epoch toggle and delta-change map in dashboard.

### S1+S2 Crop Classification — Iowa

Sentinel-1 · Sentinel-2 · scikit-learn · USDA CDL

SAR + optical fusion over 672 fields. S2-only OA 86.3% → fused 92.7% (+6.4 pp). Truth / single-sensor / fused toggle on map.

### NDVI Deforestation — Rondonia, Brazil

Google Earth Engine · Landsat 8/9 · LandTrendr

2,698 km<sup>2</sup> cleared 2015–2024. Breakpoint detection picks up the 2019 fire spike and the 2024 Lula policy reversal clearly.

### Jakarta Compound Flood Risk

rasterio · scipy.ndimage · OSMnx

4-factor risk index — HAND, slope, drainage density, imperviousness — mapped for 1,360 schools and 138 hospitals.

### 15-Minute City Accessibility — Paris

OSMnx · NetworkX · Shapely · alphashape

87.7% of hex cells reach 5+ of 6 service categories in a 15-min walk. 47,812-node network. 8 Métro anchor isochrones in dashboard.

### Wildfire Spread Simulation — Park Fire CA 2024

Rothermel model · LANDFIRE · HRRR · VIIRS

Cellular automaton with hourly HRRR wind forcing. 961/1,024 cells burned by hour 120. Peak spread 28 m/h. Validated against VIIRS fire perimeter.

### Urban Heat Island — Tokyo August 2024

Landsat 9 · Sentinel-2 · scipy

5.6 degC LST delta between Yamanote core and suburban baseline. Spearman rho = -0.82 with NDVI across a 32x32 grid.

### Rooftop Solar Potential — Lisbon

py3dep · pvlib · PVGIS · OSMnx

12.5 GWh/yr across 450 rooftops — equivalent to 4,178 Portuguese households. Terrain slope/aspect and skyline shading factored in per building.

### H3 Mobility Analytics — NYC Yellow Taxi

h3-py · DuckDB · pandas · pyarrow

57,000 daily trips in H3 hexes. Surfaces AM commute-inbound / PM airport-dropoff asymmetry across 200 hexes.

### Sentinel-5P NO<sub>2</sub> — Delhi

TROPOMI · xarray · CPCB

100% WHO exceedance in January vs 10% in June. Winter inversion vs monsoon washout clearly separated.

### **AIS Vessel Tracking — Singapore Strait**

AIS · H3 res-8 · DBSCAN · MarineTraffic

978-vessel snapshot. 200 anchored in 8 DBSCAN clusters. +18% backlog Jan–Apr 2024 tied to Red Sea diversion.

### **Coastal Erosion — Florida Atlantic 2014–2024**

Sentinel-2 · NDWI · scikit-image · Shapely

15 stations over a decade. Daytona Beach -27.4 m. Miami Beach +24.0 m from renourishment. Hurricane Irma 2017 footprint visible.

### **Cafe Location Intelligence — Tbilisi**

OSMnx · GeoPandas · scipy · Leaflet

62,849-node pedestrian network. 8,415 hex cells over 504 km<sup>2</sup>. Spearman rho = 0.39 vs real cafe distribution.

### **Advanced VRP — Tbilisi**

OSMnx · NetworkX · OR-Tools

Greedy vs Clarke-Wright vs OR-Tools GLS over 60 stops / 5 vehicles. OR-Tools won by 20.3% on total drive time.

### **Web GIS Dashboard — Delhi PS**

Leaflet.js · Chroma.js · vanilla JavaScript

Single-file dashboard, ~480 lines of vanilla JS, no framework, no build step. 6 presets, split view, live histogram, URL-hash state persistence.

---

## **EDUCATION**

### **Bachelor of Science, Geodesy and Geoinformatics Engineering**

Georgian Technical University | Tbilisi, Georgia | 2022 – 2026 (Expected)

- Core coursework: geodesy, satellite positioning systems, remote sensing, GIS analysis, cartography, and spatial data management.
- Parallel to the degree: designed and deployed 17 satellite analysis projects covering InSAR, ML classification, routing, mobility, wildfire, solar, and climate domains — all live on GitHub Pages with interactive dashboards.
- Freelance GIS and software engineering work throughout the programme.